PRICE STATISTICS

The consumer price index: from COICOP 2018 to online prices

Definition, basket, compilation methods, and a hands-on lab in Côte d'Ivoire and Senegal

Three key messages

The CPI is a policy instrument before it is a calculation



A high-stakes indicator

The CPI guides monetary policy, indexes wages and contracts, and deflates the national accounts. A one-point error is paid for everywhere.



It all starts with the basket

What we measure depends on what goes into the basket and how much weight it gets. COICOP 2018 structures the basket across WAEMU since the 2023 base.



New sources

In our lab, online prices and prices published by institutions cover 7 of 13 divisions. A useful complement, not a substitute.

Acronyms

Institutions and technical terms used in this presentation

CPI	Consumer price index
HICP	Harmonised index of consumer prices (European Union)
WAEMU	West African Economic and Monetary Union
ECB	European Central Bank
ANSD	National Agency of Statistics and Demography (Senegal)
ANARE-CI	National electricity sector regulatory authority (Côte d'Ivoire)
ARTP	Telecommunications and postal regulatory authority (Senegal)
HCPI	Harmonised consumer price index (WAEMU)
COICOP	Classification of Individual Consumption According to Purpose (United Nations)
BCEAO	Central Bank of West African States
EHCVM	Harmonised survey on household living conditions (WAEMU)
OCPV	Food products marketing support office (Côte d'Ivoire)
CRSE	Energy sector regulatory commission (Senegal)
NPISHs	Non-profit institutions serving households

01

Understanding the CPI

What does the CPI really measure, and what is it for?



What is the consumer price index?

A short definition, and four scope choices that make it operational

The CPI measures the average change, between two periods, in the prices of a fixed basket of goods and services **acquired by households for consumption**, at constant quality.

Who?

Reference population: all resident households, or a group (urban, wage earners).

Where?

Geographic coverage: capital, main cities or whole country, urban and rural.

What?

Consumer goods and services, classified by COICOP. Not investment, not savings.

Which prices?

Purchase prices actually paid, taxes included, including discounts and administered prices.

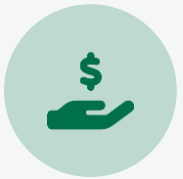
One indicator, five uses that drive decisions

The variety of uses is what demands high standards of quality and transparency



Monetary policy

The BCEAO sets its rates based on WAEMU inflation, measured by the HCPI.



Indexation

Minimum wages, pensions, rents and contracts are adjusted in line with the CPI.



National accounts

Price indices deflate consumption to obtain volumes.



Multilateral surveillance

The WAEMU inflation criterion is read from each member state's HCPI.



Purchasing power

Public debate, wage bargaining, cost-of-living measurement.

HCPI, HICP, deflator: cousins, not twins

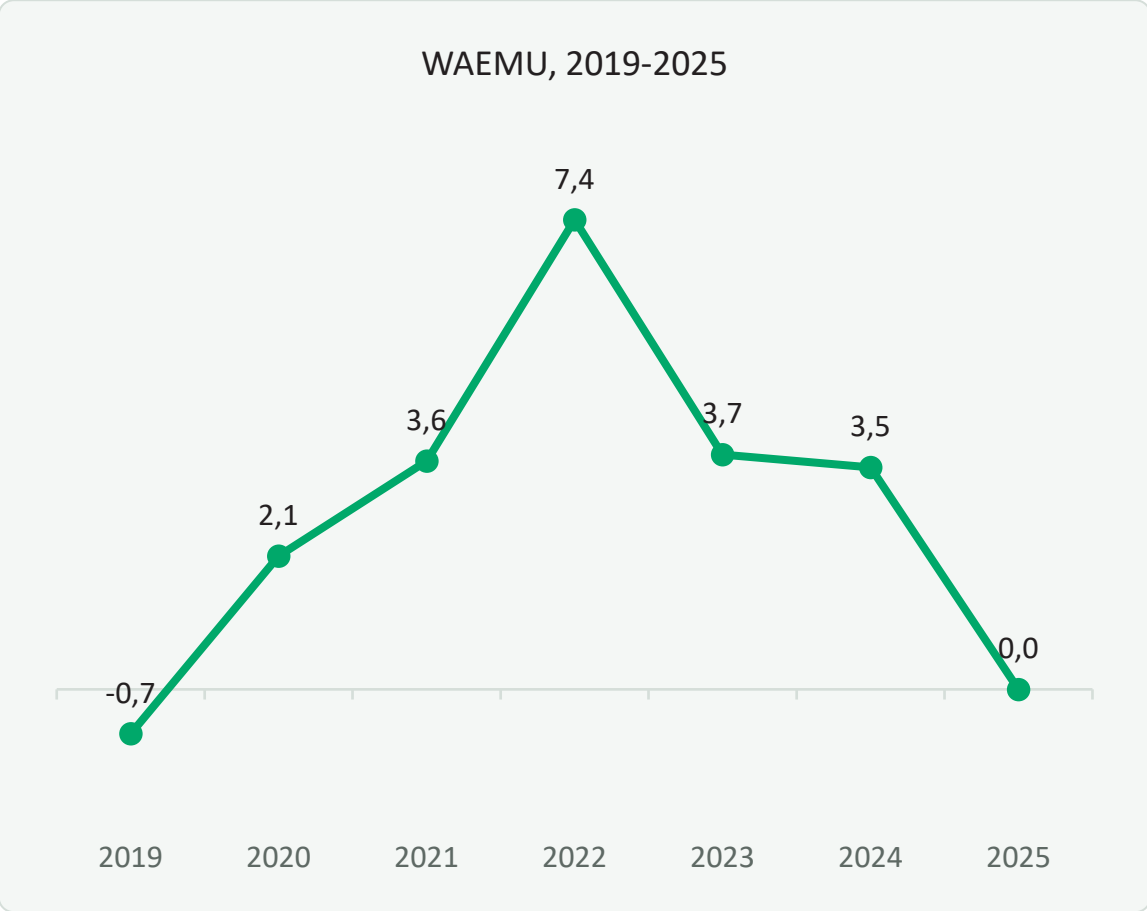
Three measures of consumer prices, three frameworks, three uses

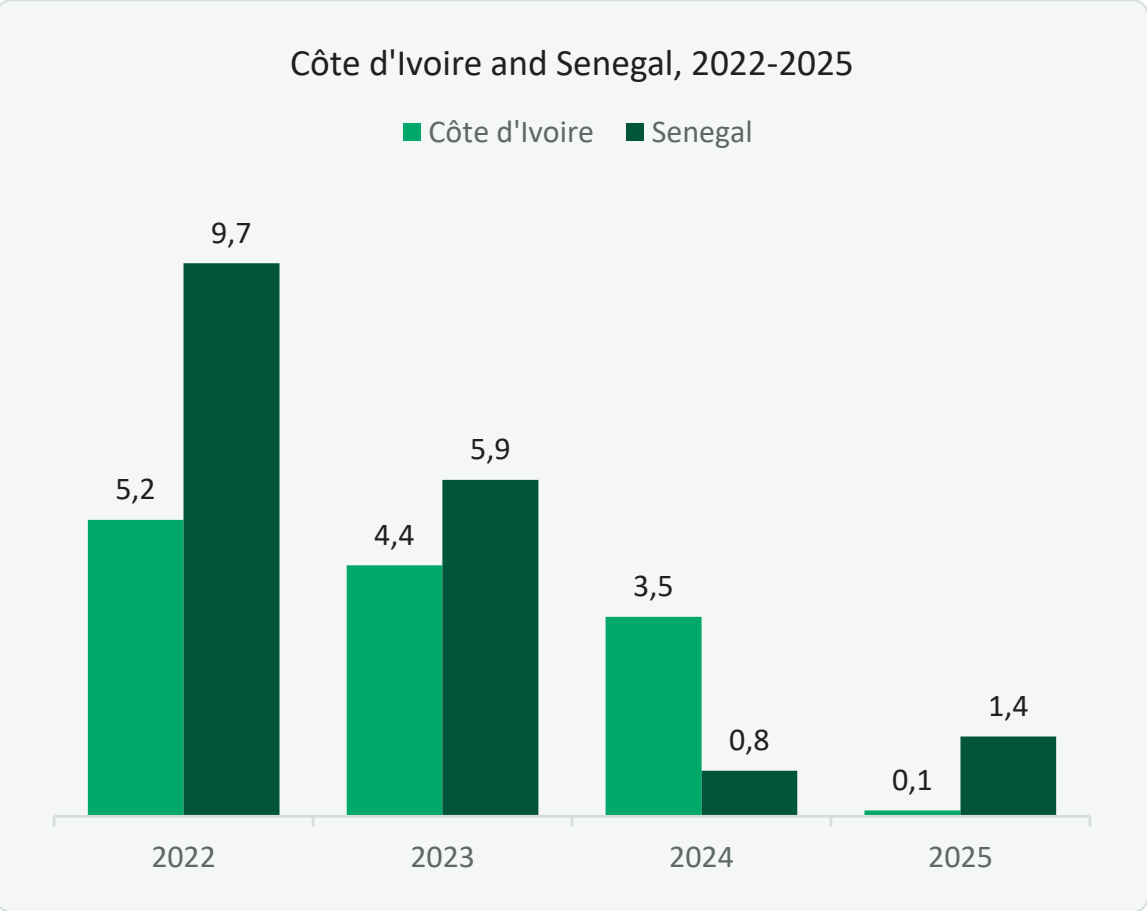
	HCPI (WAEMU)	HICP (European Union)	Consumption deflator
What it is	The national CPI of each member state, harmonised across the eight countries	Harmonised index of EU countries, published by Eurostat	Ratio of consumption at current prices to volume, in the national accounts
Classification	Classification derived from COICOP 2018, 13 divisions, base 2023	ECOICOP version 2, identical to COICOP 2018, since January 2026	COICOP 2018 in EU national accounts since 2024
Weights	EHCVM 2021-2022 household survey	National accounts, updated every year	Change every period (Paasche-type formula)
Main use	BCEAO monetary policy, convergence criterion	ECB 2% inflation target	Macroeconomic analysis, consumption volumes

Sources: BCEAO, annual report on consumer prices in WAEMU in 2025; ANSD, HCPI March 2026; Eurostat; ECB.

WAEMU: from the 2022 peak to stable prices in 2025

Annual average inflation measured by the HCPI, %





Source: BCEAO, Annual report on consumer prices in WAEMU in 2025 (June 2026), annex 1 and table 1. Annual averages.

02

The basket and COICOP 2018

What goes into the basket, with what weight, and under which framework?



The basket, a snapshot of consumption

From household budgets to items priced each month: the example of Senegal's HCPI, base 2023

Budget survey

EHCVM 2021-2022: what households spend, item by item

Classification

Spending coded in a classification derived from COICOP 2018: 13 divisions

Weights

Share of each item in total spending, out of 10,000

Items

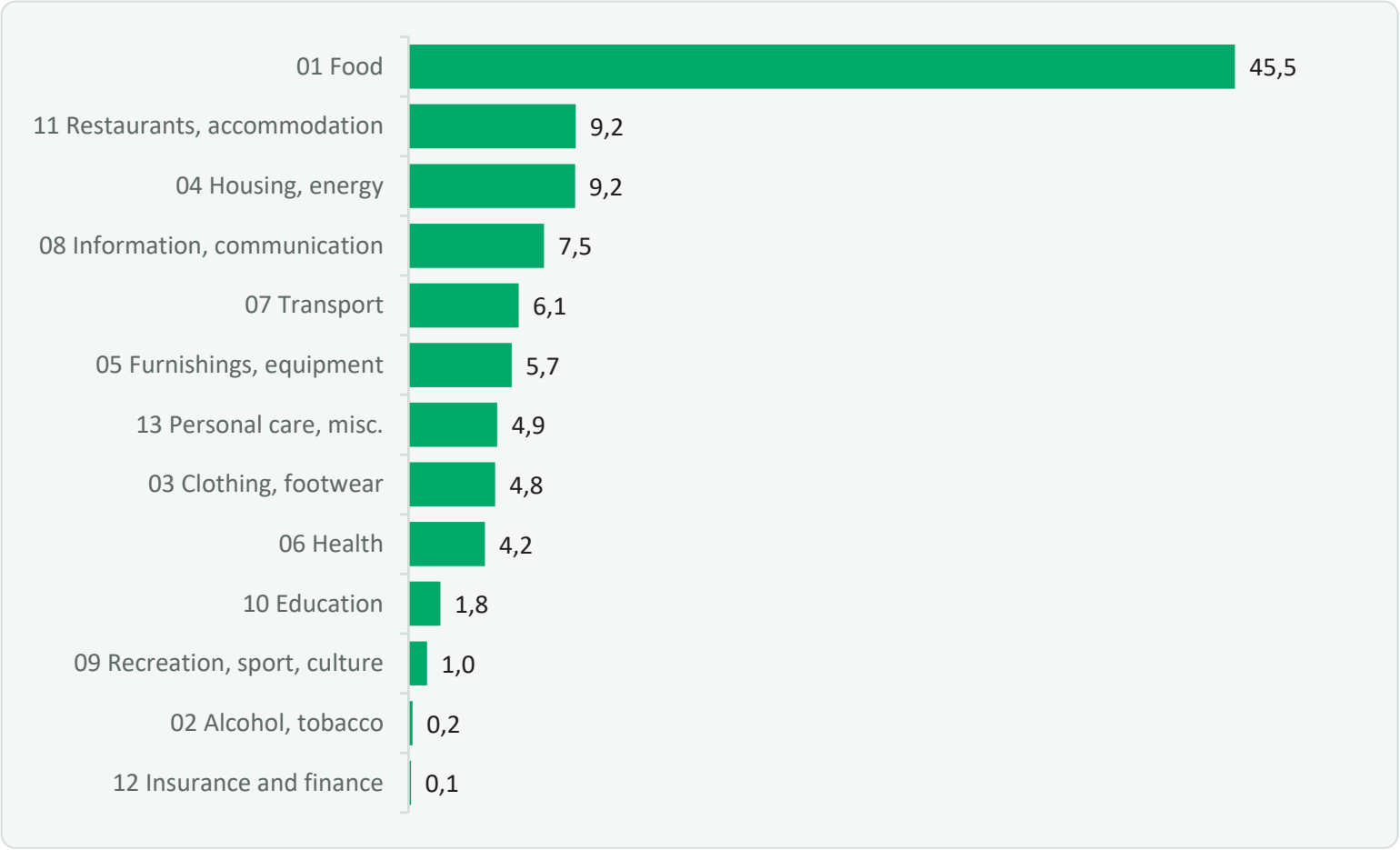
Specific products priced each month: "imported broken rice, 1 kg"

Three rules for a good basket

- Representative: selected items reflect what households really buy, including at open-air markets.
- Stable but updated: the basket is fixed between two rebasings, then renewed to follow consumption.
- Traceable: each item is linked to a COICOP subclass, which ensures international comparability.

Weights: a highly concentrated basket

Weight of the 13 divisions in Senegal's HCPI, base 2023, % of spending



45.5% for food: almost half of the basket

64% in three divisions: food, restaurants, housing-energy

3% for the four smallest: insurance, alcohol-tobacco, recreation, education

Source: ANSD, HCPI March 2026, table 1 (weights out of 10,000, converted to %).

COICOP 2018, the framework behind the basket

Four nested levels and a product type for each subclass

01 — **Division** 15 items

Food and non-alcoholic beverages

01.1 — **Group** 63 items

Food

01.1.4 — **Class** 186 items

Milk, other dairy products and eggs

01.1.4.2 — **Subclass** **338 items · new in 2018**

Skimmed milk

Subclass product type

ND — **Non-durable goods**

bread, milk, fuel

SD — **Semi-durable goods**

clothing, small appliances

D — **Durable goods**

car, refrigerator

S — **Services**

rent, transport, repair

The CPI covers divisions 01 to 13; divisions 14 and 15 cover individual consumption by NPISHs and general government.

What COICOP 2018 changes for a CPI

Services better identified, and a European migration that left headline inflation intact

Goods and services separated

Repair, rental and installation get their own service items.

Delivery identified

Separately invoiced delivery fees and parcels become distinct services.

Digital division 08

IT hardware and software join information and communication.

Dedicated division 12

Insurance and financial services leave "miscellaneous goods and services".

Lessons learned: the January 2026 HICP

≤ 0.1 pp — difference in euro area headline inflation

1.2 pp — maximum revision to unprocessed food

≈ 1% — weight of games of chance, a new HICP item

Sources: United Nations (UNSD), COICOP 2018; ECB, Economic Bulletin 3/2026, box "What's new in the HICP?".

03

Compiling the index

How do thousands of price quotes become a single published index?

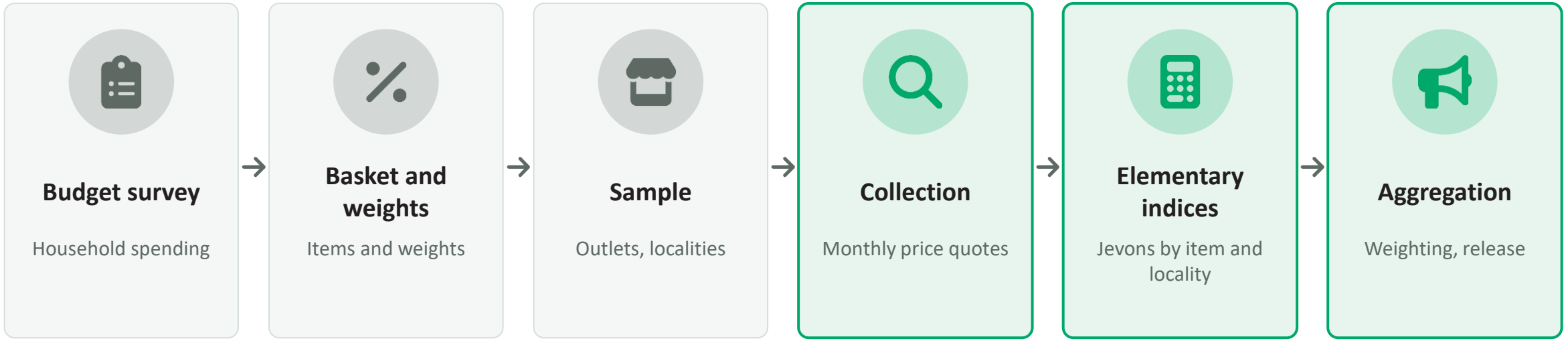


The CPI production chain

Two rhythms: a base set every five to ten years, production repeated every month

EVERY 5 TO 10 YEARS · REBASING

EVERY MONTH · PRODUCTION



Checks at every stage: outlier prices, missing items, replacements, consistency across localities and with the previous month. Discrepancies are documented before release.

Collecting prices: four complementary sources

Field collection remains the backbone; digital sources extend it

Field collection

Price collectors record prices at markets and in shops, on a fixed schedule.

+ Covers informal and rural markets

– Costly, limited sample

Administered prices

Tariffs set and published: electricity, water, fuel, regulated transport.

+ Comprehensive and official

– Rarely change: a revision is easily missed

Scanner data

Supermarket transactions: prices and quantities actually sold.

+ Huge volumes, known quantities

– Agreements to negotiate, large retailers only

Online prices

Automated collection of prices shown on retail websites.

+ Daily, low-cost, very detailed

– Posted prices, urban buyers, frequent discounts

Elementary indices: why choose Jevons

Worked example: two products on alternating discount, whose prices return to their starting level

Price (CFA francs)	Month 0	Month 1	Month 2
Product A	100	50	100
Product B	100	200	100

Jevons

geometric mean of price relatives

100 → 100 → 100

Dutot

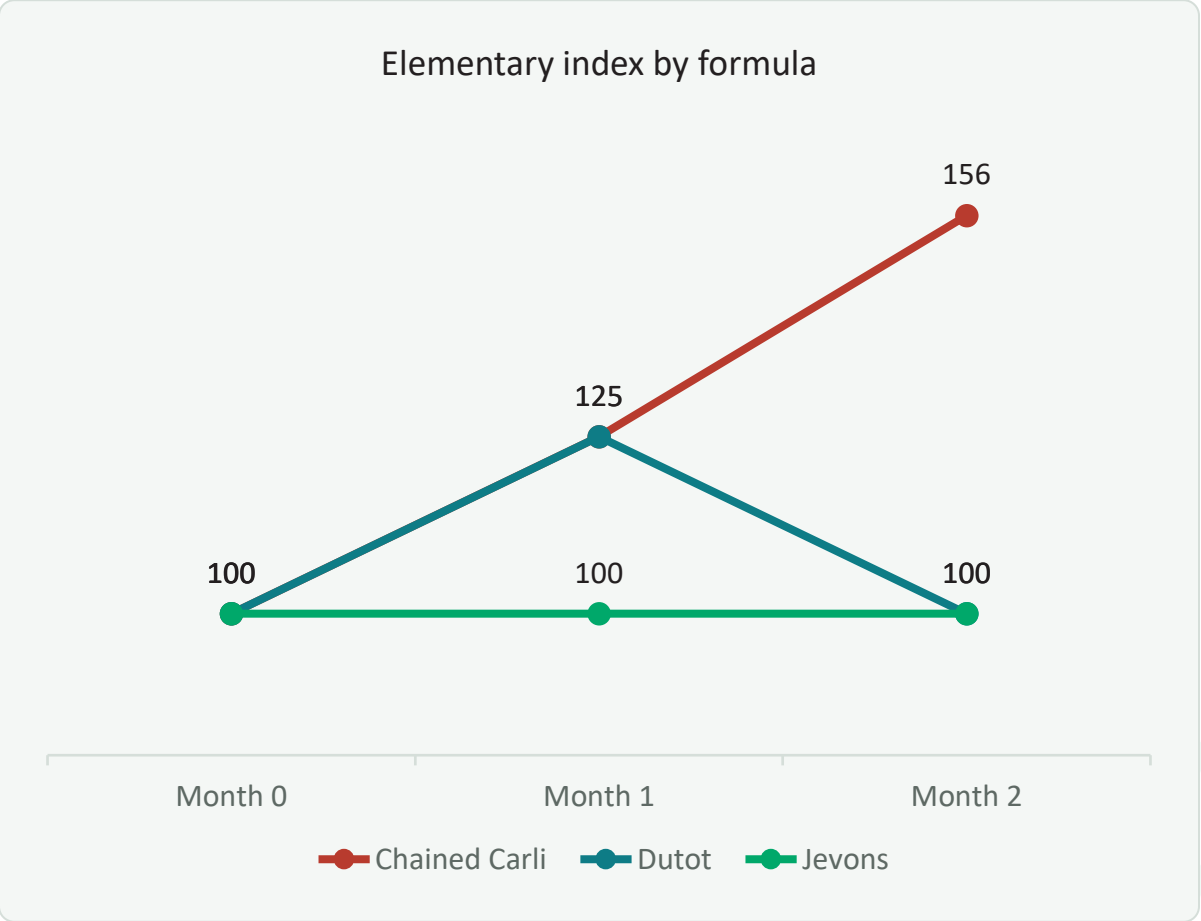
ratio of average prices

100 → 125 → 100

Chained Carli

arithmetic mean of price relatives

100 → 125 → 156



Illustrative example. Reference: Consumer Price Index Manual (2020), chapter on elementary indices.

Aggregating, chaining, rebasing

From elementary indices to the all-items index released every month

$$I_t = \Sigma\, w_i \times I_{i,t}$$

The all-items index is the average of elementary indices I_i, weighted by expenditure shares w_i from the budget survey.

Young or Laspeyres formula

Weights come from the budget survey, earlier than the price base period. ANSD uses the Young formula for Senegal's HCPI.

Chaining

Weights are renewed at each rebasing, and series are linked to stay continuous over time.

Rebasing

Every 5 to 10 years: new survey, new basket, new classification. In WAEMU, the base is 2023 = 100.

Four challenges that define CPI quality

Measuring at constant quality when products, seasons and discounts change

Missing products

Out of stock, outlet closed: impute the price from comparable products rather than carrying it forward.

Replacements and quality

One model replaces another: separate the price rise from the quality gain, for example with hedonic methods.

Seasonal products

Fruit and vegetables missing part of the year: variable baskets or seasonal imputation.

Discounts

Struck-through prices make series bounce: several quotes per month, geometric mean, and the Jevons formula.

04

An online price index

Tested in Côte d'Ivoire: what can online prices and prices published by institutions bring to a CPI?



Architecture: collect on site, analyse anywhere

Collection and compilation are separate: only a file travels between them

ON THE OFFICE WORKSTATION

1 **Jumia collection**

Online prices, robots.txt respected, crawler identified

2 **Institutional sources**

Electricity tariffs, market prices, telecoms index



.zip archive: Parquet, Excel, manifest

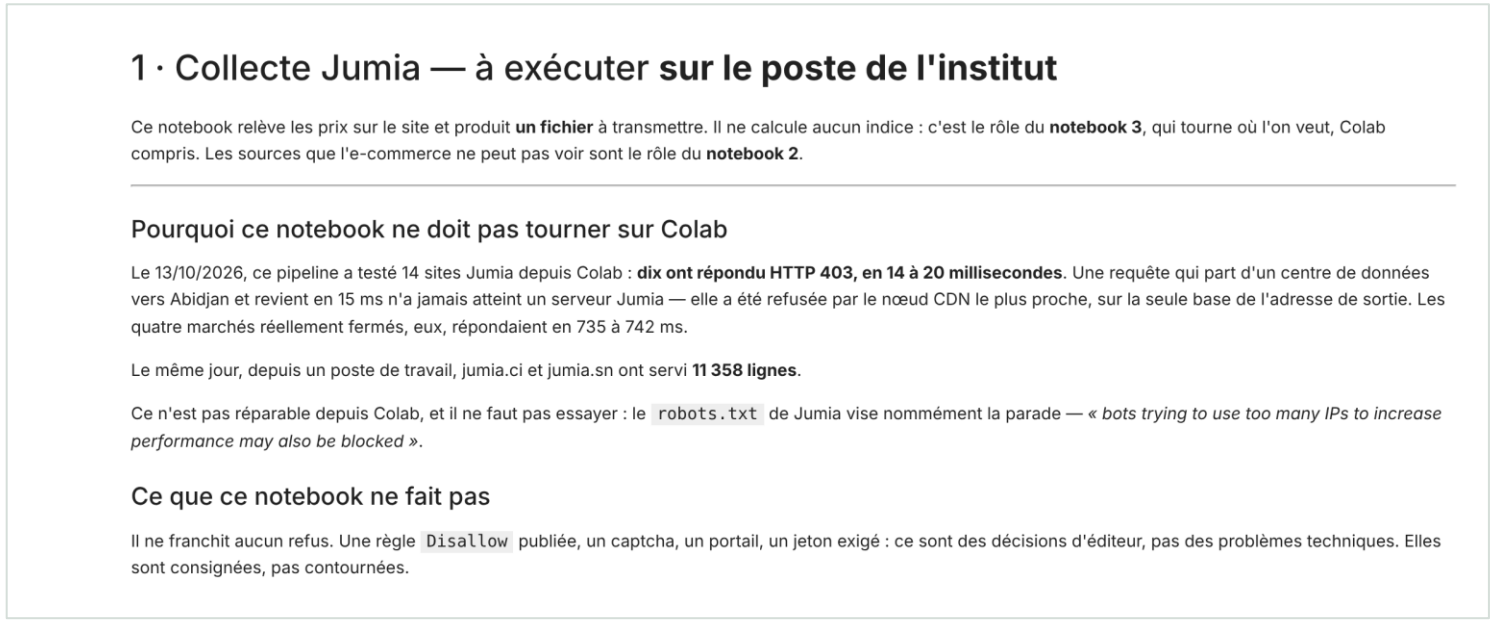
ON COLAB OR ON SITE

3 **Analysis**

- Merges and checks the archives
- Classifies products by COICOP
- Compares classification methods
- Weights and computes Jevons, Carli, GEKS
- Produces a standalone HTML report

Notebook 1: collecting online prices without forcing anything

Notebook screenshot (in French): collection must run from the office workstation



1 · Collecte Jumia — à exécuter **sur le poste de l'institut**

Ce notebook relève les prix sur le site et produit **un fichier** à transmettre. Il ne calcule aucun indice : c'est le rôle du **notebook 3**, qui tourne où l'on veut, Colab compris. Les sources que l'e-commerce ne peut pas voir sont le rôle du **notebook 2**.

Pourquoi ce notebook ne doit pas tourner sur Colab

Le 13/10/2026, ce pipeline a testé 14 sites Jumia depuis Colab : **dix ont répondu HTTP 403, en 14 à 20 millisecondes**. Une requête qui part d'un centre de données vers Abidjan et revient en 15 ms n'a jamais atteint un serveur Jumia — elle a été refusée par le nœud CDN le plus proche, sur la seule base de l'adresse de sortie. Les quatre marchés réellement fermés, eux, répondaient en 735 à 742 ms.

Le même jour, depuis un poste de travail, jumia.ci et jumia.sn ont servi **11 358 lignes**.

Ce n'est pas réparable depuis Colab, et il ne faut pas essayer : le `robots.txt` de Jumia vise nommément la parade — *« bots trying to use too many IPs to increase performance may also be blocked »*.

Ce que ce notebook ne fait pas

Il ne franchit aucun refus. Une règle `Disallow` publiée, un captcha, un portail, un jeton exigé : ce sont des décisions d'éditeur, pas des problèmes techniques. Elles sont consignées, pas contournées.

10 / 14

Jumia sites refused from Colab, in 14 to 20 ms: blocked by the content delivery network (CDN), not by Jumia

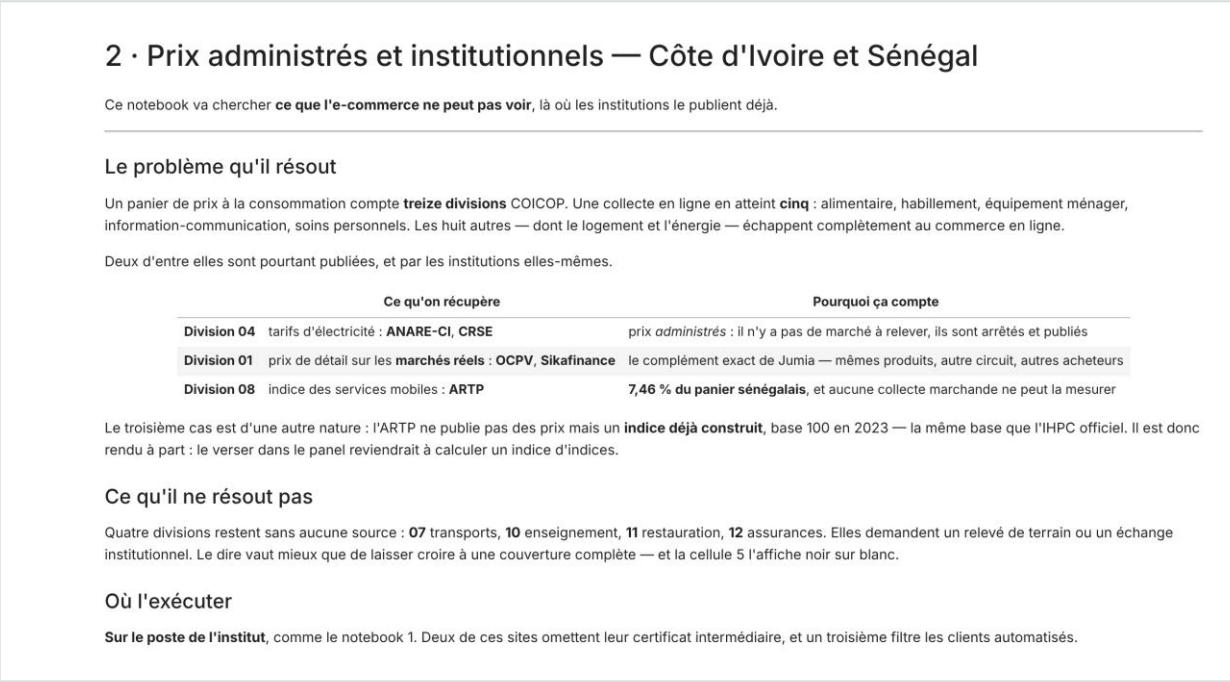
11,358

rows served the same day by jumia.ci and jumia.sn to a workstation

- The crawler identifies itself with a contact address
- robots.txt and the delay between requests are respected
- A refusal is logged, never bypassed

Notebook 2: reaching what e-commerce cannot see

Administered and market prices already published by Ivorian and Senegalese institutions



2 · Prix administrés et institutionnels — Côte d'Ivoire et Sénégal

Ce notebook va chercher **ce que l'e-commerce ne peut pas voir**, là où les institutions le publient déjà.

Le problème qu'il résout

Un panier de prix à la consommation compte **treize divisions** COICOP. Une collecte en ligne en atteint **cinq** : alimentaire, habillement, équipement ménager, information-communication, soins personnels. Les huit autres — dont le logement et l'énergie — échappent complètement au commerce en ligne.

Deux d'entre elles sont pourtant publiées, et par les institutions elles-mêmes.

	Ce qu'on récupère	Pourquoi ça compte
Division 04	tarifs d'électricité : **ANARE-CI**, **CRSE**	prix *administrés* : il n'y a pas de marché à relever, ils sont arrêtés et publiés
Division 01	prix de détail sur les **marchés réels** : **OCPV**, **Sikafinance**	le complément exact de Jumia — mêmes produits, autre circuit, autres acheteurs
Division 08	indice des services mobiles : **ARTP**	**7,46 % du panier sénégalais**, et aucune collecte marchande ne peut la mesurer

Le troisième cas est d'une autre nature : l'ARTP ne publie pas des prix mais un **indice déjà construit**, base 100 en 2023 — la même base que l'IHPC officiel. Il est donc rendu à part : le verser dans le panel reviendrait à calculer un indice d'indices.

Ce qu'il ne résout pas

Quatre divisions restent sans aucune source : **07** transports, **10** enseignement, **11** restauration, **12** assurances. Elles demandent un relevé de terrain ou un échange institutionnel. Le dire vaut mieux que de laisser croire à une couverture complète — et la cellule 5 l'affiche noir sur blanc.

Où l'exécuter

Sur le poste de l'institut, comme le notebook 1. Deux de ces sites omettent leur certificat intermédiaire, et un troisième filtre les clients automatisés.

04

Electricity

ANARE-CI and CRSE tariffs: administered prices, set and published

01

Physical markets

OCPV and Sikafinance retail prices: the exact complement to Jumia

08

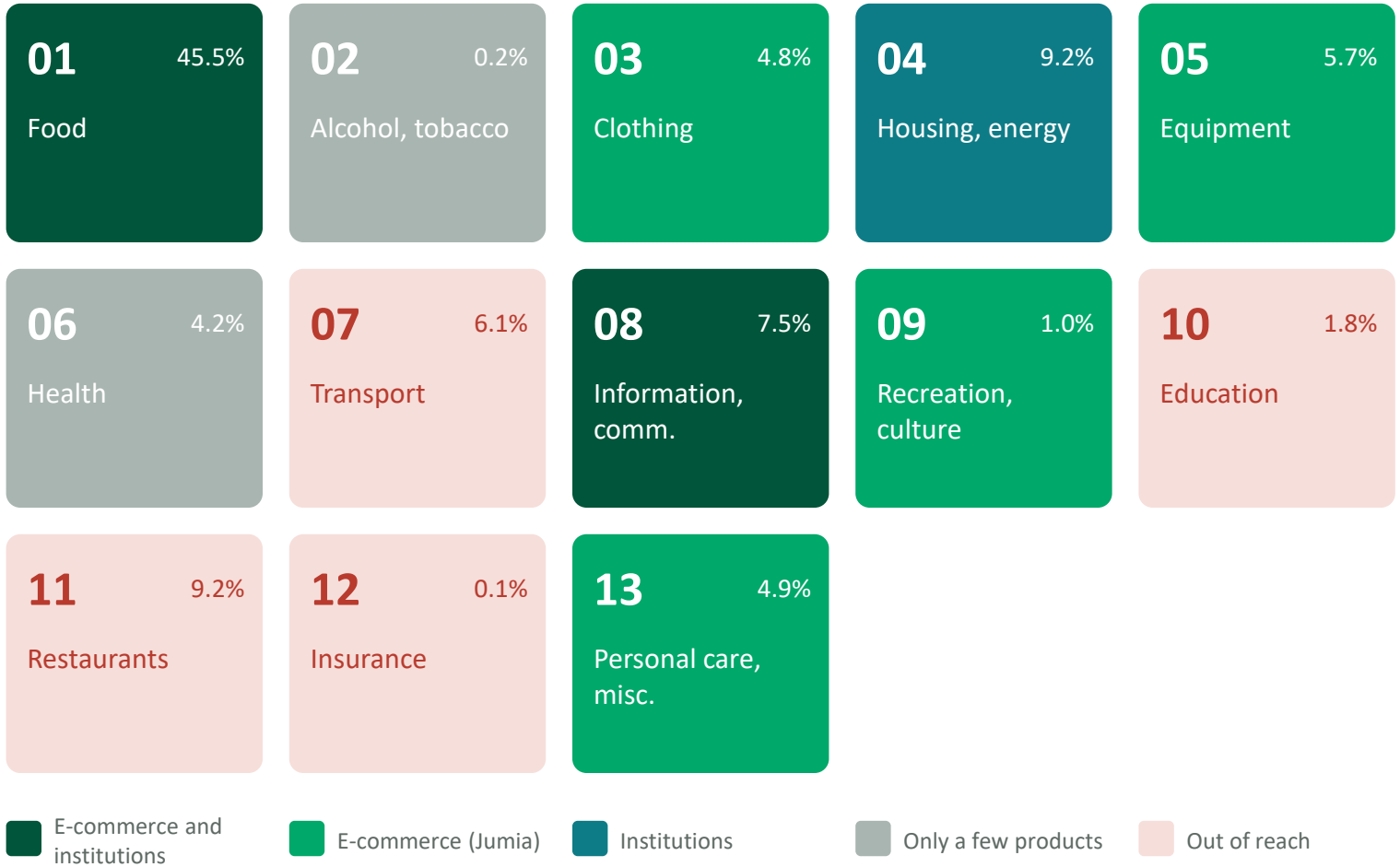
Mobile services

ARTP index, base 2023: 7.46% of Senegal's basket, kept separate

⚠ A ready-made index is not mixed with prices: that would compute an index of indices.

Coverage: 7 of 13 divisions, 78% of the weight

Divisions reached by the 14 October 2026 collection, with each one's weight in Senegal's HCPI (illustration)



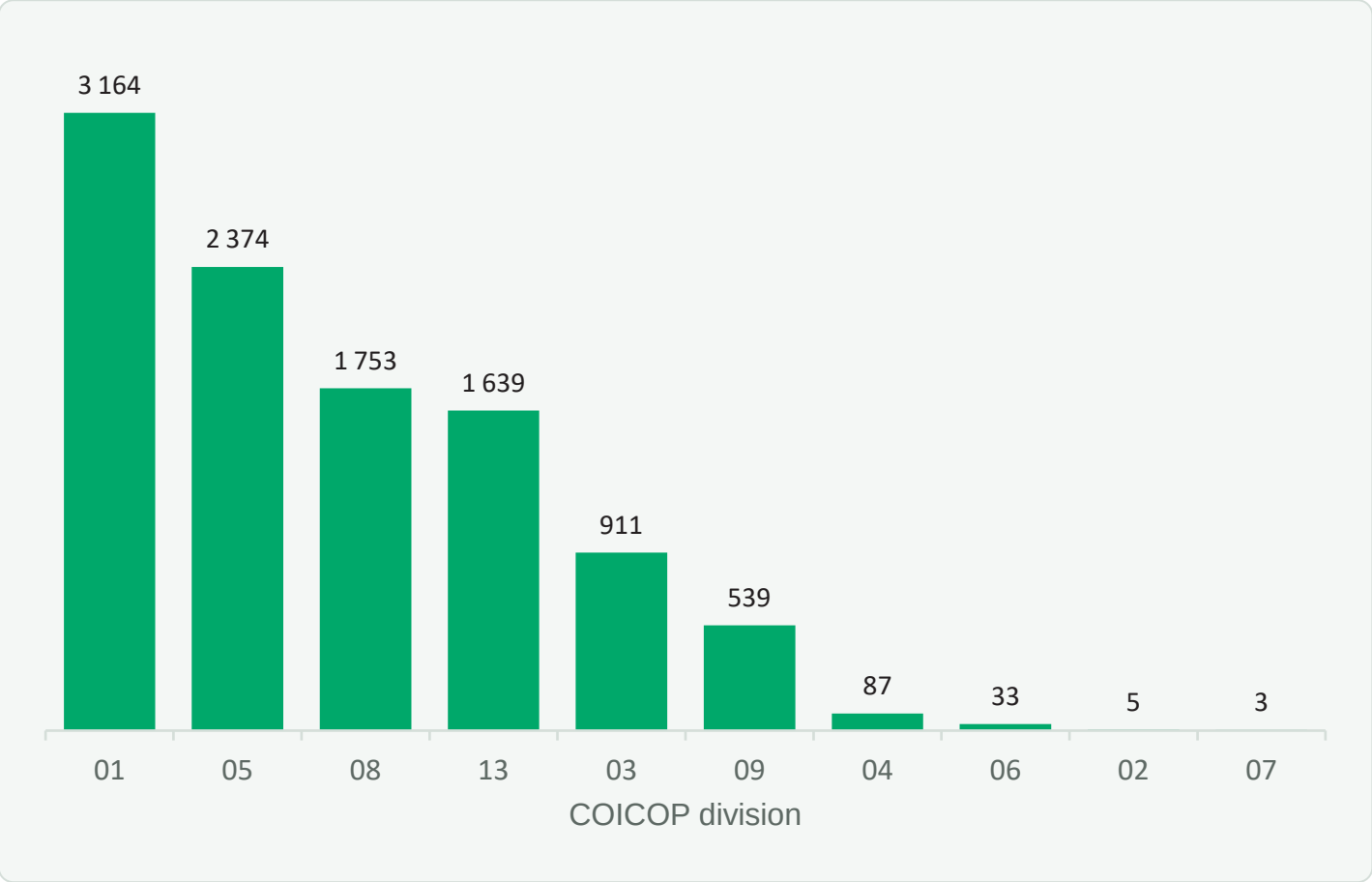
78% of the basket weight reached by the 7 covered divisions

17% in 4 divisions out of reach: transport, education, restaurants, insurance

4% in 2 divisions barely touched: alcohol-tobacco and health

First collection: 10,523 price quotes in one day

Côte d'Ivoire, 14 October 2026: distinct products by COICOP division after classification



7,759 Jumia quotes (notebook 1)

2,764 institutional quotes: OCPV, Sikafinance, tariffs (notebook 2)

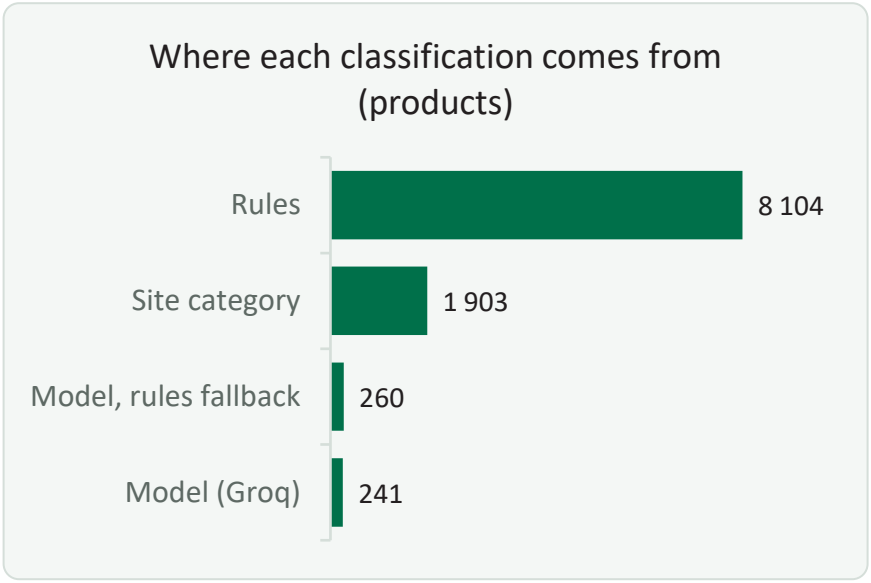
15 duplicates detected and removed when merging

100% of the 10,508 products assigned to a COICOP class

Source: notebook 3 run (jumia-cpi 1.0.0), archives panel_CI_2026-10-14_1331.zip and officiel_CI_2026-10-14_1418.zip.

Classifying by COICOP: rules, model and benchmark

A method is not chosen in advance: it is measured on a set whose correct answer is known



Jeu **synthétique** : contrôle de la chaîne de mesure, pas comparaison de méthodes. Les règles y obtiennent près de 100 % parce qu'elles partagent s vocabulaire.

380 produits annotés, 29 classes

méthode	couverture	justesse_globale	justesse_sur_classés	précision_macro	rappel_macro	F1_macro	F1_pondéré	secondes	produits_par_seconde
regles	0.99	0.99	1.00	1.00	0.99	0.99	0.99	0.05	7 008.60
llm	1.00	0.96	0.96	0.92	0.92	0.92	0.96	214.57	1.80
mixte	1.00	0.96	0.96	0.92	0.92	0.92	0.96	106.50	3.60

Meilleur F1 macro : **regles** à 0.994, avec 98.7% de couverture et 100.0% de justesse sur les produits classés.

99% rules F1 on the synthetic set: a pipeline check, not proof

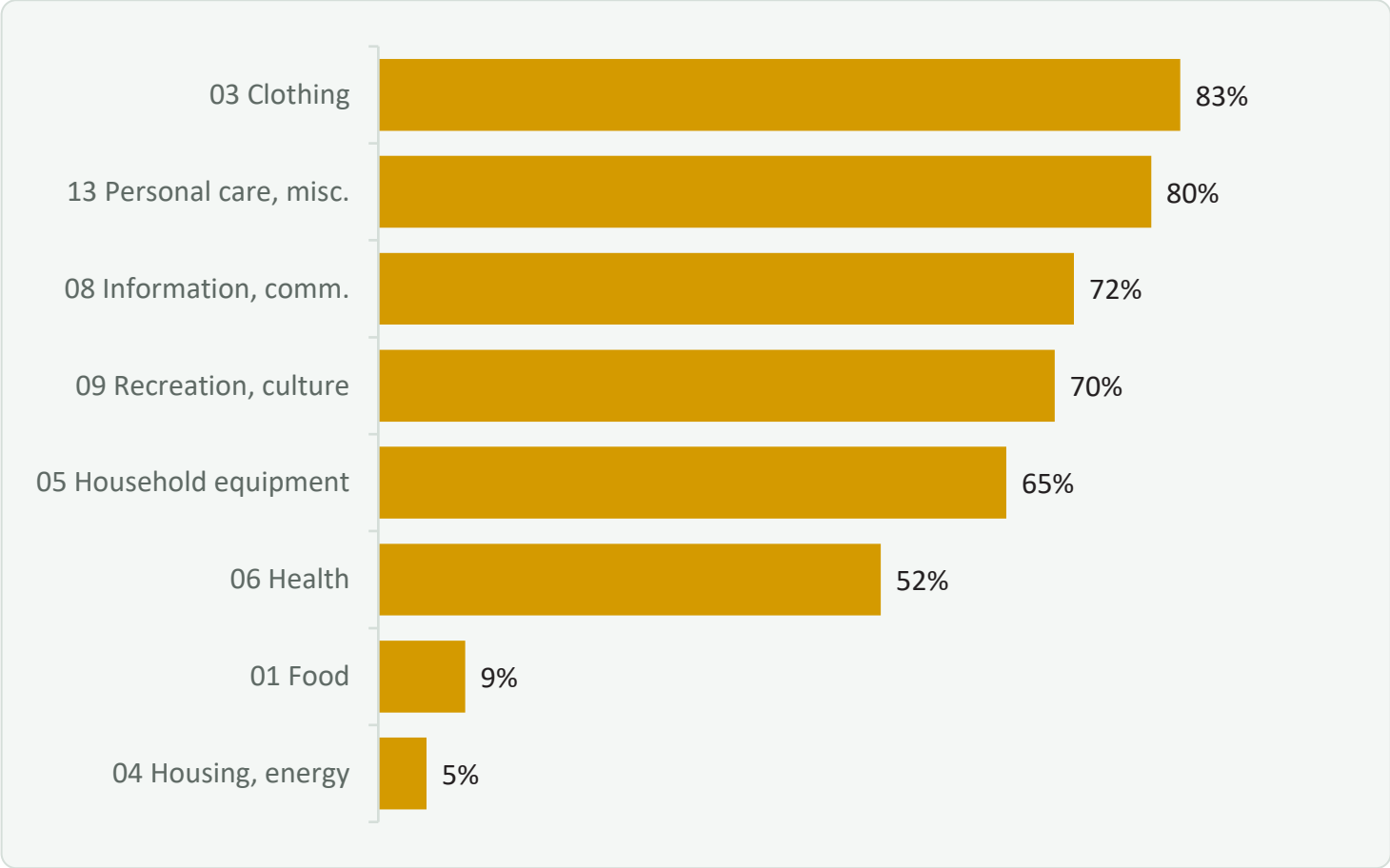
92% language model macro F1, with transport / housing confusions

× 3,894 rules speed compared with the model: hours saved on 10,000 products

Source: notebook 3, benchmark on 380 annotated products and 29 classes (demo set). 94 real products are ready to be annotated for a conclusive benchmark.

Discounts: the specific challenge of online prices

Share of products showing a struck-through price, by division, October 2026 collection in Côte d'Ivoire



83%

of clothing items show a struck-through price, against 9% for food

What the lab does

- Computes the index on posted prices and on regular prices
- Recommends three quotes per month, combined by geometric mean
- Uses Jevons, and shows the Carli drift

Source: notebook 3, figure "Frequency of struck-through prices by division and month". Divisions 02 and 07 excluded (fewer than 10 products).

What this index measures, and what it does not

A complement to the official CPI, to be presented with its limits

Urban, connected buyers : open-air markets and the informal sector are covered only through the OCPV

Mostly goods : housing, transport, health and education remain barely or not covered

Posted, not paid, prices : no quantities sold; discounts weigh heavily

Fast-moving catalogue : 2,900 new products in October; new items only enter the following month

Before releasing a figure

- The institute's weights, not equal shares
- File with a provenance manifest
- Not the demo data set
- Benchmark on your own data
- At least two months of quotes

Status on 14 October 2026: a single collection, so the index stands at 100. A second collection, the following month, is needed to measure a change.

Testing the lab in six steps

From setup to report; collection is repeated every month at the same time

01 Set up

Run preparer.bat in the jumia-cpi folder: it creates a dedicated environment.

02 Collect online

Notebook 1 on the workstation: choose the country, fill in CONTACT_EMAIL.

03 Complete

Notebook 2: administered and market prices for Côte d'Ivoire or Senegal.

04 Share

Share the .zip archives, never the Excel file alone: it can retype codes.

05 Analyse

Notebook 3 on Colab: official weights, benchmark on your data.

06 Publish

Download rapport.html: charts included, it opens anywhere.

05

Challenges and conclusion

What challenges remain for measuring prices reliably?



Current challenges for the CPI

Four debates every CPI compiler should be able to explain

Measurement bias

Product substitution, new products, quality effects, outlet changes: the CPI tends to slightly overstate inflation.

Owner-occupied housing

Owner-occupiers' housing costs are not yet in the European HICP; approaches are still debated.

E-commerce

A growing share of purchases, very volatile prices, and delivery fees now identified by COICOP 2018.

Perceived inflation

Frequent purchases, low-income households more exposed to food: the gap with the average index fuels distrust.

In summary

A reliable CPI rests on a well-built basket, a documented method and diverse sources

01

The CPI drives decisions

Monetary policy, indexation, national accounts: measurement quality has a real cost.

02

COICOP 2018 structures the basket

Thirteen divisions, services better identified, a 2023 base shared across WAEMU.

03

New sources complement

78% of the weight reached in the lab, but four divisions still require field collection.

Next steps

- Second collection in November
- Official weights for Côte d'Ivoire
- Annotate the 94 benchmark products
- Add rents (division 04)
- Compare with the official HCPI

Sources

Reference documents and data used in this presentation

Consumer Price Index Manual (2020)

Concepts and Methods. Joint publication of the international statistical organisations.

imf.org

BCEAO (2026)

Annual report on consumer prices in WAEMU in 2025 (in French).

bceao.int

ANSD (2026)

Harmonised consumer price index, March 2026: base-2023 weights and methodological note (in French).

ansd.sn

United Nations, UNSD (2018)

Classification of Individual Consumption According to Purpose (COICOP) 2018.

unstats.un.org/unsd/classifications

ECB (2026)

What's new in the HICP? The 2026 classification update. Economic Bulletin 3/2026.

ecb.europa.eu

jumia-cpi 1.0.0 lab (2026)

Collection and analysis notebooks; run of 14 October 2026 in Côte d'Ivoire.

notebooks 1, 2 and 3

THE CONSUMER PRICE INDEX

Thank you

Measuring prices rigorously, and opening the CPI to new data sources